AI-1 Spring 26 Project Proposal

Team Members

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Idea Short Name (mind maze)

Idea Description

AI Maze Adventure is a first-person 3D maze exploration game built entirely in Python using the Pygame library. It combines three sophisticated systems into one cohesive experience: • A procedurally generated maze rendered in pseudo-3D using raycasting (the same technique used in Wolfenstein 3D and the original DOOM). • An AI-powered pathfinding system using the A* algorithm that guides the player toward the maze exit. • A fully interactive 8-puzzle (sliding tile puzzle) mini-game that the player must solve to earn AI hints.

Agent Architecture

☐ Simple Reflex Agent

√ Model-Based Reflex Agent

√ Goal-Based Agent

☐ Utility-Based Agent

Problem-Solving Framework

√ Classical Search

☐ Local Search (Optimization)

☐ Adversarial Search (Two-Player Games)

☐ Constraint Satisfaction Problem

Proposed Algorithm(s)

1-Recursive Backtracking — Maze Generation : Category: Depth-First Search with randomization. Produces a 'perfect maze' — exactly one path between any two cells. No loops, no isolated sections **2-A* is a best-first search algorithm that finds the shortest path in a weighted graph. It maintains a priority queue sorted by f(n) = g(n) + h(n)** **3-xManhattan Distance between two points (r1,c1) and (r2,c2) = |r1-r2| + |c1-c2|.** **4-DDA Raycasting Algorithm**

PEAS

Performance Measures	1- Reach the maze goal using the **shortest path** 2- Minimize number of steps taken 3- Reduce wrong decisions at intersections 4- Solve the puzzle in minimum moves and time
Environment	1- 3D Maze (partially observable from player view) 2-Static walls and paths 3-Intersections (decision points) 4- Puzzle environment (3×3 N-Puzzle)
Actuators	1- Move Forward / Backward 2-Turn Left / Right 3- Request AI Hint 4-Interact with Puzzle (move tiles)
Sensors	1-Player position in the maze 2-Detection of walls and available paths 3-Detection of intersections 4-Current puzzle state (tile positions)

Environment Properties

☐ Fully Observable √ Partially Observable	√ Deterministic ☐ Stochastic ☐ Strategic	√ Discrete ☐ Continuous
√ Single-Agent ☐ Multi-Agent	√ Sequential ☐ Episodic	√ Static ☐ Dynamic